

Deep Learning for Antibiotic Resistance Prediction in *Klebsiella* from Petri Dish Images

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1 Motivation

Antibiotic resistance is a pressing global health challenge, and *Klebsiella* species represent one of the major pathogens of concern. Traditional methods for identifying bacterial subtypes and assessing resistance rely on labor-intensive laboratory assays that require expert interpretation. Automating this process through computer vision and machine learning could accelerate diagnosis and improve reproducibility.

Petri dish images provide a direct and widely available representation of bacterial growth. However, they pose significant challenges: colonies overlap, illumination and contrast vary across images, and subtle morphological differences can determine clinically relevant subtypes. To address these issues, we propose a deep learning framework that integrates segmentation and classification, with the possibility of enriching predictions through clinical metadata.

2 Segmentation of Bacterial Colonies

The first step of the proposed framework is to design a robust segmentation system capable of accurately isolating *Klebsiella* colonies in Petri dish images. Traditional segmentation techniques often struggle with overlapping colonies, varying illumination, and heterogeneous colony morphology. To address these challenges, we draw inspiration from the approach presented by Hickl et al. [1], which takes advantage of synthetic data generation, domain adaptation, and advanced segmentation models to achieve state-of-the-art performance in dense and multi-species bacterial colony segmentation.

2.1 Synthetic Data Generation

A key bottleneck in training colony segmentation models is the scarcity of annotated datasets. Manual labeling of colonies is labor-intensive and error-prone, particularly when colonies are dense or overlapping. To overcome this limitation, we propose generating synthetic Petri dish images together with ground-truth segmentation masks. Colony shapes can be modeled using simple geometric primitives or stochastic growth simulations, while realistic agar backgrounds can be overlaid to mimic experimental conditions. Each synthetic image is paired with its corresponding ground-truth mask, enabling the creation of arbitrarily large annotated datasets without manual intervention.

2.2 Style Transfer with CycleGAN

Although synthetic images provide perfectly aligned ground-truth masks, their visual appearance often differs significantly from real Petri dish photographs. To bridge this domain gap, we adopt a CycleGAN-based style transfer module. The CycleGAN is trained to translate synthetic images into realistic counterparts, capturing lighting conditions, textural patterns, and subtle

visual cues present in actual clinical data. This step ensures that the synthetic dataset more closely resembles the target domain while preserving its segmentation annotations, making it suitable for training robust models.

2.3 Training the Segmentation Model

With a large corpus of realistic synthetic data available, we train an advanced segmentation network such as **Omnipose**, a model specifically designed to handle the irregular shapes and dense packing typical of bacterial colonies. Omnipose learns to predict colony boundaries even under challenging scenarios, such as overlapping growth or heterogeneous morphology. Training on CycleGAN-enhanced synthetic images allows the model to generalize more effectively to real-world Petri dish images, reducing the dependency on manually annotated datasets.

2.4 Application to Real Images

Once trained, the segmentation model is applied directly to real Petri dish images. Colonies are automatically detected and segmented, providing clean crops that can be passed to the downstream classification network. This approach ensures that even in difficult cases—such as plates with many overlapping colonies or suboptimal imaging conditions—segmentation quality remains sufficiently high to support accurate subtype classification. Furthermore, because the segmentation pipeline is based on synthetic data generation, it can be adapted to new experimental setups or imaging modalities with minimal additional labeling effort.

The segmentation output is a set of colony crops that serve as inputs for the classification stage.

3 Colony Classification with Deep Networks

Each segmented colony image is passed through a convolutional neural network (CNN) to predict its bacterial subtype. Architectures such as ResNet are strong candidates due to their balance of accuracy and efficiency. The model outputs a probability distribution over possible subtypes.

To obtain a prediction at the plate level, we employ a majority-voting strategy: the most frequently predicted subtype among colonies on a plate is assigned as the final decision. This reduces sensitivity to misclassifications of individual colonies and increases robustness.

4 Integration of Metadata

Beyond image analysis, we consider incorporating clinical and experimental metadata, such as:

- Type of culture medium used in the assay.
- Patient-specific information (e.g., age, prior antibiotic exposure).

These additional features can be fused with image-derived embeddings through multimodal architectures, such as:

- Concatenation of learned colony representations with metadata vectors.
- Attention-based fusion, allowing the model to weigh metadata relevance dynamically.

This integration may improve predictive performance and provide insights into how external factors influence bacterial resistance.

5 Challenges and Open Questions

Several aspects require systematic investigation:

5.1 Segmentation Quality

The accuracy of downstream classification is highly dependent on segmentation performance. A key open question is whether errors in colony extraction significantly bias subtype prediction, and how robust the classifier can be to imperfect segmentation.

5.2 Colony-Level vs. Plate-Level Labels

While colony-level classification provides granularity, clinical relevance often resides at the plate level. Strategies for aggregating colony predictions into plate-level decisions (e.g., majority voting, probabilistic aggregation) need to be compared.

5.3 Model Architecture

CNNs such as ResNet provide a strong baseline, but more recent architectures (e.g., EfficientNet, Vision Transformers) may improve performance. The trade-off between accuracy, interpretability, and computational cost must be considered.

5.4 Metadata Fusion

The impact of metadata on prediction accuracy is uncertain. Evaluating whether multimodal integration yields significant gains over image-only models is essential. Moreover, metadata use introduces challenges in data harmonization and missing-value handling.

6 Expected Outcomes

The proposed framework aims to:

- Automate colony detection and classification in Petri dish images.
- Provide robust plate-level predictions of *Klebsiella* subtypes.
- Assess the role of metadata in enhancing predictive models of antibiotic resistance.

Ultimately, this research seeks to contribute to faster, more reliable, and interpretable tools for microbiological diagnostics.

References

- [1] V. Hickl, A. Khan, R. M. Rossi, B. F. Silva, K. Maniura-Weber, Segmentation of dense and multi-species bacterial colonies using models trained on synthetic microscopy images, PLOS Computational Biology 21 (4) (2025) e1012874.